

Homework 3: Coordinated Movement

Due date: November 7, 2025, 11:59 PM on Gradescope

Problem 1: Leader–Follower Consensus with Multiple Leaders

In this problem, we will consider what happens when a group has multiple leaders in a linear consensus protocol. This model serves as a simple abstraction for understanding how agents synchronize to a shared reference when multiple leaders broadcast potentially different reference signals. Each node in the network is associated with a scalar state variable $x_i(t) \in \mathbb{R}$, representing, for instance, a position, opinion, or phase. The leaders are sometimes referred to as *pinning* nodes, as they seek to “pin” or synchronize the behavior of the rest of the network to their own state. We will analyze how the followers’ states evolve and converge depending on the number and consistency of the leaders.

Let $\mathcal{G}_F = (V_F, E_F)$ denote the undirected and connected communication graph among the N_F followers. For any pair $(i, j) \in E_F$, the symmetric adjacency weight is $a_{ij} = 1$, and $a_{ij} = 0$ otherwise. The leaders are indexed by $V_L = \{1, \dots, N_L\}$, with their states being reference signals $r_\ell(t)$, $\ell \in V_L$. Whenever follower i receives information from leader ℓ , we set the leader-follower adjacency weight $b_{i\ell} = 1$, and $b_{i\ell} = 0$ otherwise. Let $\mathcal{N}_i^F$ denote the set of neighboring followers that influence follower i , and $\mathcal{N}_i^L$ the set of leaders that influence follower i .

The continuous-time consensus protocol for the followers is given by

$$\dot{x}_i = -\kappa_F \sum_{j \in \mathcal{N}_i^F} a_{ij}(x_i - x_j) - \kappa_L \sum_{\ell \in \mathcal{N}_i^L} b_{i\ell}(x_i - r_\ell(t)), \quad i \in V_F, \quad (1)$$

where $\kappa_F > 0$ and $\kappa_L > 0$ are the coupling gains between followers and leaders, respectively. In vector form, the follower dynamics can be written compactly as

$$\dot{x}_F = -(\kappa_F L_F + \kappa_L D_{\text{pin}})x_F + \kappa_L B r(t), \quad (2)$$

where L_F is the Laplacian of the follower graph, $D_{\text{pin}} = \text{diag}(d_i^{\text{pin}})$ with $d_i^{\text{pin}} = \sum_{\ell} b_{i\ell}$, and $B \in \mathbb{R}^{N_F \times N_L}$ is the leader–follower incidence matrix with entries $b_{i\ell}$. The vector $r(t) = [r_1(t), \dots, r_{N_L}(t)]^\top$ collects all leader reference signals.

1. **(15 points) Identical Leaders.** Assume that all leaders share the same constant reference $r(t) \equiv r$. Define the error variables $e_i = x_i - r$, and collect them into the vector $e = [e_1, \dots, e_{N_F}]^\top$. Show that the error dynamics reduce to

$$\dot{e} = -(\kappa_F L_F + \kappa_L D_{\text{pin}})e. \quad (3)$$

Consider the function

$$V(e) = \frac{1}{2} e^\top e. \quad (4)$$

Verify that V is positive definite, and therefore a valid candidate Lyapunov function. Compute $\dot{V}$ along the error system trajectories and show that

$$\dot{V} = -e^\top (\kappa_F L_F + \kappa_L D_{\text{pin}})e \leq -\lambda_{\min}(\kappa_F L_F + \kappa_L D_{\text{pin}}) \|e\|^2. \quad (5)$$

Using this result, show that the follower group opinions converge to the leader reference, with rate at least

$$\alpha = \kappa_F \lambda_2(L_F) + \kappa_L d_{\min}^{\text{pin}}. \quad (6)$$

(Hint: for a dynamical system $\dot{x} = f(x)$ with Lyapunov function $V(x)$, if $\dot{V}(x) \leq -\alpha \|x\|^2$ the origin is exponentially stable with rate at least α). Briefly discuss how increasing the number of leaders, changing placement of leaders in the graph, and the strength of leader-follower coupling κ_L affects convergence to the reference.

2. **(15 points) Heterogeneous Leaders.** Now suppose that the leaders hold distinct static references $r_\ell(t) = r_\ell \in \mathbb{R}$, $\ell = 1, \dots, N_L$. Show that the equilibrium follower state x_F^* satisfies

$$(\kappa_F L_F + \kappa_L D_{\text{pin}})x_F^* = \kappa_L B r. \quad (7)$$

Interpret the equilibrium opinions x_F^* of the followers related to the reference states r of the leaders. How does the steady-state opinion profile depend on the number of leaders and the structure of the leader-follower connections? To investigate this question, write down an expression for the matrix $M \in \mathbb{R}^{N_F \times N_L}$ such that $x_F^* = M r$ and compute M numerically for several different choices of $\mathcal{G}_F$, B , N_L , and κ_L for $N_F = 5$. Interpret the resulting equilibrium expression. Do you expect the opinions of agents to converge exactly the state of one of the leaders or to some other quantity? What tunes how spread out the final opinions are? Does the relative influence of different leaders change based on the different choices you made in initializing the system?

3. **(20 points) Simulation Comparison** Simulate (1) for $N_F = 20$ followers for three scenarios:

- (a) $N_L = 1$;
- (b) $N_L = 3$ with identical leaders;
- (c) $N_L = 3$ with heterogeneous leaders.

For each scenario, use both an undirected ring graph and a random (Erdős-Rényi) graph for the follower network (Hint: you can use the `erdos_renyi_graph` function in the `networkx` package in Python https://networkx.org/documentation/stable/reference/generated/networkx.generators.random_graphs.erdos_renyi_graph.html to generate the graph; choose edge creation probability p sufficiently high to make sure the connectivity assumption is satisfied). Plot the follower states $x_i(t)$ over time, and discuss how the convergence rate and the final steady-state configuration change with the number of leaders and the strength of κ_F, κ_L . Investigate the following questions:

- i. Is there an advantage to (b) over (a) - how do the convergence rates compare?
- ii. For (b) and (c) over a random graph, does the placement of the leader nodes change the outcome of the consensus process? Connect this to what you learned in Part 2 of the problem.

Problem 2: Flocking

Flocking is a type of cohesive, decentralized movement in groups of agents. In this problem, we will compare and contrast three related algorithms for flocking motion: the original Boids model of Reynolds that synthesizes velocity alignment, cohesion, and separation forces; the Olfati-Saber flocking algorithm with a trajectory reference; and a Model Predictive Control optimal flocking algorithm.

Problem setup. We consider N point-mass agents in $\mathbb{R}^2$ with double-integrator dynamics

$$\dot{p}_i = v_i, \quad \dot{v}_i = u_i, \quad i = 1, \dots, N, \quad (8)$$

where $p_i, v_i, u_i \in \mathbb{R}^2$ are the position, velocity, and control input of agent i . For a sensing radius $R > 0$ define the time-varying proximity neighborhood $\mathcal{N}_i(R) := \{j \neq i : \|p_i - p_j\| \leq R\}$. To streamline your algorithm implementations you may wish to write a function that, given the position vectors of N agents, outputs the unweighted proximity-based graph adjacency matrix with a specified sensing radius R .

1. **(20 points) Reynolds Boids (alignment, cohesion, separation).** Implement a continuous-time

Boids field for each agent i consisting of:

$$\begin{aligned} u_i^{\text{sep}} &= k_{\text{sep}} \sum_{j \in \mathcal{N}_i^R} \phi_{\text{rep}}(\|p_i - p_j\|) \frac{p_i - p_j}{\|p_i - p_j\|}, \\ u_i^{\text{ali}} &= k_{\text{ali}} (\bar{v}_i - v_i), \quad \bar{v}_i = \frac{1}{|\mathcal{N}_i^R|} \sum_{j \in \mathcal{N}_i^R} v_j, \\ u_i^{\text{coh}} &= k_{\text{coh}} (\bar{p}_i - p_i), \quad \bar{p}_i = \frac{1}{|\mathcal{N}_i^R|} \sum_{j \in \mathcal{N}_i^R} p_j, \end{aligned}$$

with neighborhood $\mathcal{N}_i^R = \{j \neq i : \|p_i - p_j\| \leq R\}$, repulsion shaping $\phi_{\text{rep}}(r) = \max\{0, r_0 - r\}$, and nonnegative gains $k_{\text{sep}}, k_{\text{ali}}, k_{\text{coh}}$. The total control is $u_i = u_i^{\text{sep}} + u_i^{\text{ali}} + u_i^{\text{coh}}$. Simulate with $N \in [20, 50]$ with random position and velocity initial conditions. Plot trajectories of the agents over time. Do the following: (i) plot connectivity of the proximity graph over time (plot $\lambda_2(L)$, the second smallest eigenvalue of the graph Laplacian, over time); (ii) number of connected components (Hint: the number of connected components is equal to the multiplicity of the zero eigenvalue in the unweighted graph Laplacian). Report whether the group fragments into several sub-flocks or remains cohesive as the number of agents N grows.

2. (30 points) Olfati–Saber Flocking Algorithm 2 with global reference.

We now implement a flocking algorithm with a virtual leader proposed by Olfati-Saber. The control has three terms

$$u_i = u_i^{\text{int}} + u_i^{\text{ali}} + u_i^\gamma,$$

where u_i^{int} is a gradient term from a smooth collective potential that enforces cohesion and separation, u_i^{ali} is a velocity alignment term, and u_i^γ is the navigational feedback toward a virtual leader (the global reference signal).

Setup. Let $\varepsilon > 0$ and define the σ -norm for $z \in \mathbb{R}^2$ by

$$\|z\|_\sigma = \frac{1}{\varepsilon} (\sqrt{1 + \varepsilon \|z\|^2} - 1), \quad \nabla_z \|z\|_\sigma = \frac{z}{\sqrt{1 + \varepsilon \|z\|^2}}.$$

Define the bump function $\rho_h : [0, \infty) \rightarrow [0, 1]$ with parameter $h \in (0, 1]$ by

$$\rho_h(s) = \begin{cases} 1, & 0 \leq s < h, \\ \frac{1}{2} \left[1 + \cos\left(\pi \frac{s-h}{1-h}\right) \right], & h \leq s \leq 1, \\ 0, & s > 1, \end{cases}$$

and the state-dependent adjacency weights

$$a_{ij}(p) = \rho_h\left(\frac{\|p_i - p_j\|}{R}\right).$$

Pairwise potential and interaction term. To construct a smooth pairwise potential with a finite cut-off, we integrate an *action function* $\phi_\alpha(z)$ that vanishes for all $z \geq R$. Define

$$\phi_\alpha(z) = \rho_h\left(\frac{z}{R}\right) \phi(z - d), \quad \phi(z) = \frac{1}{2}[(a + b)\sigma_1(z + c) + (a - b)],$$

where

$$\sigma_1(z) = \frac{z}{\sqrt{1 + z^2}},$$

and $\phi(z)$ is an uneven sigmoidal function with parameters satisfying $0 < a \leq b$ and $c = \frac{|a-b|}{\sqrt{4ab}}$ to guarantee $\phi(0) = 0$. The function $\rho_h(s)$ is the smooth bump function

$$\rho_h(s) = \begin{cases} 1, & 0 \leq s < h, \\ \frac{1}{2} \left[1 + \cos\left(\pi \frac{s-h}{1-h}\right) \right], & h \leq s \leq 1, \\ 0, & s > 1, \end{cases}$$

which enforces a finite sensing range R .

With the σ -distance $z_{ij} = \|p_i - p_j\|_\sigma$, the total potential and corresponding interaction term are

$$U(p) = \frac{1}{2} \sum_i \sum_{j \in \mathcal{N}_i} \phi_\alpha(z_{ij}), \quad u_i^{\text{int}} = - \sum_{j \in \mathcal{N}_i} \phi_\alpha(z_{ij}) \nabla_{p_i} z_{ij},$$

where $\nabla_{p_i} z_{ij} = \frac{p_i - p_j}{\sqrt{1 + \varepsilon \|p_i - p_j\|^2}}$. This formulation yields smooth, bounded attractive and repulsive forces with a unique equilibrium spacing d_α and a finite interaction radius R .

Velocity alignment term. Let $L(p)$ be the graph Laplacian of the proximity graph with weights $a_{ij}(p)$. The multi-dimensional velocity consensus term is

$$u_i^{\text{ali}} = - \sum_{j \in \mathcal{N}_i} a_{ij}(p) (v_i - v_j),$$

i.e., the standard consensus protocol with state-dependent weights.

Navigational feedback (virtual leader). Let the virtual leader (“ γ -agent”) have state (p_γ, v_γ) with either constant velocity or a given trajectory. The navigational feedback is linear in position/velocity errors:

$$u_i^\gamma = -c_1(p_i - p_\gamma) - c_2(v_i - v_\gamma),$$

with gains $c_1, c_2 > 0$. In this formulation all agents are “informed,” i.e., they know (p_γ, v_γ) .

Simulations. Simulate the flocking algorithm from the same initial conditions as your simulations in part 1; pick $\varepsilon=0.1$, d such that $d < R$, $h \in [0.2, 0.5]$. Consider two scenarios: tracking of a stationary reference $p_\gamma = v_\gamma = 0$ and tracking of a constant velocity $v_\gamma \in \mathbb{R}^2$ and $p_\gamma = v_\gamma t$. Plot the trajectories in space over time. Do the following: (i) plot connectivity of the proximity graph over time (plot $\lambda_2(L)$, the second smallest eigenvalue of the unweighted graph Laplacian, over time); (ii) number of connected components (Hint: the number of connected components is equal to the multiplicity of the zero eigenvalue in the unweighted graph Laplacian); (iii) plot the average error over time between velocities of agents and the reference velocity v_γ . Compare performance of this flocking algorithm to the original Reynolds flocking from Part 1.